

Solution Architecture

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

The Rising Waters architecture follows a simple layered design: a Presentation Layer for user interaction, an Application Layer for request handling and inference, and a Data Layer for the trained model and dataset.

Layered Architecture (top to bottom)

Layer	Contents
Presentation / Client Layer	Web Browser rendering the HTML form (templates/index.html, static/style.css)
Application Layer	Flask app (app.py) — routes "/" and "/predict"; calls predict.py
ML Inference Layer	predict.py — loads models/flood_model.pkl (Joblib) and returns a prediction
Data / Storage Layer	data/flood_dataset.xlsx (training data) and models/flood_model.pkl (trained model)

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects weather inputs from the user and displays the prediction result.	HTML, CSS
Application Layer (Flask App)	Handles HTTP requests, validates/converts form inputs, and returns rendered results.	Python, Flask
ML Inference Module	Loads the trained Random Forest model and generates a flood/no-flood prediction.	Scikit-learn, Joblib, NumPy
Data Layer	Stores the historical training dataset and the serialized trained model.	Excel (openpyxl), Joblib (.pkl)